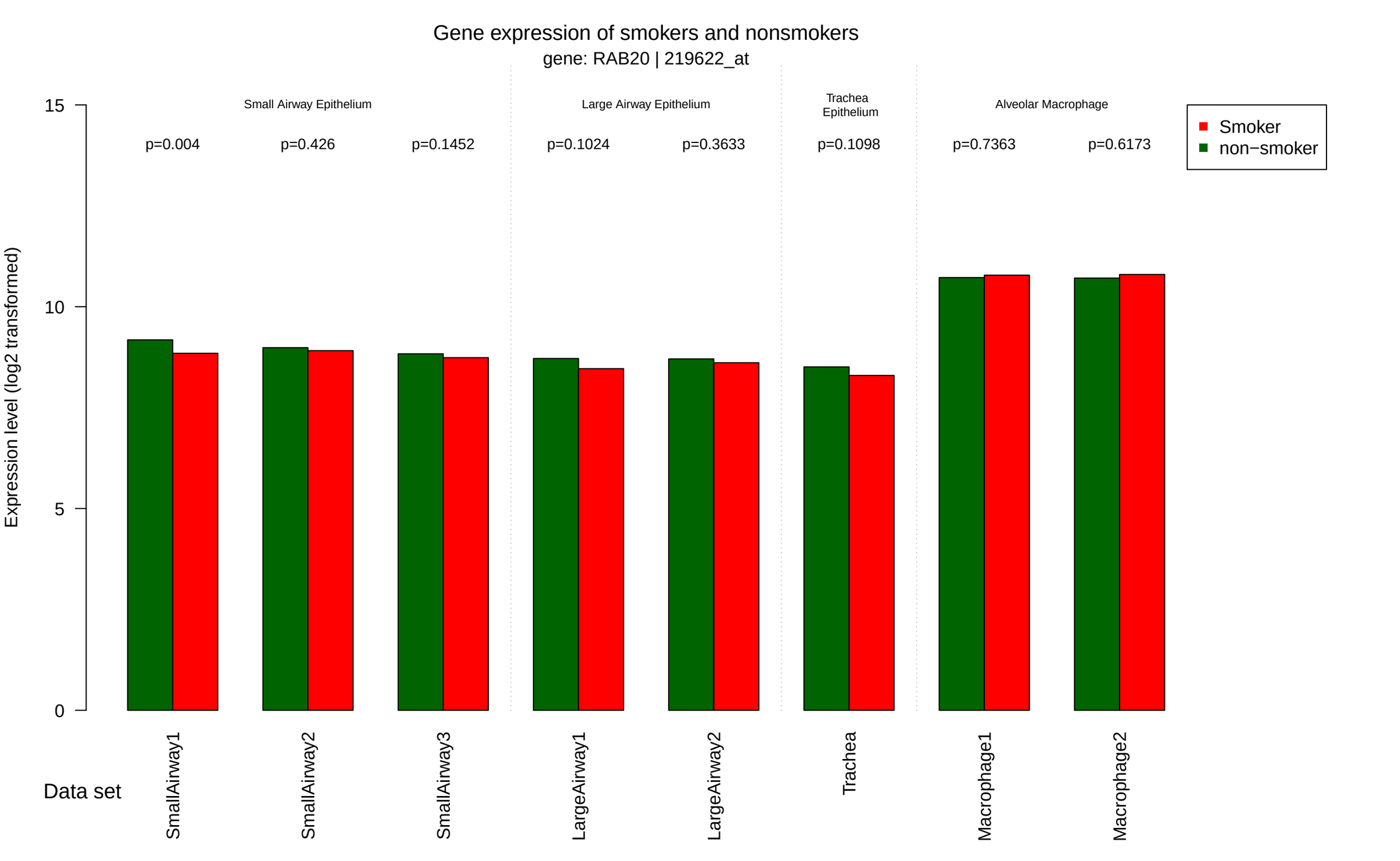
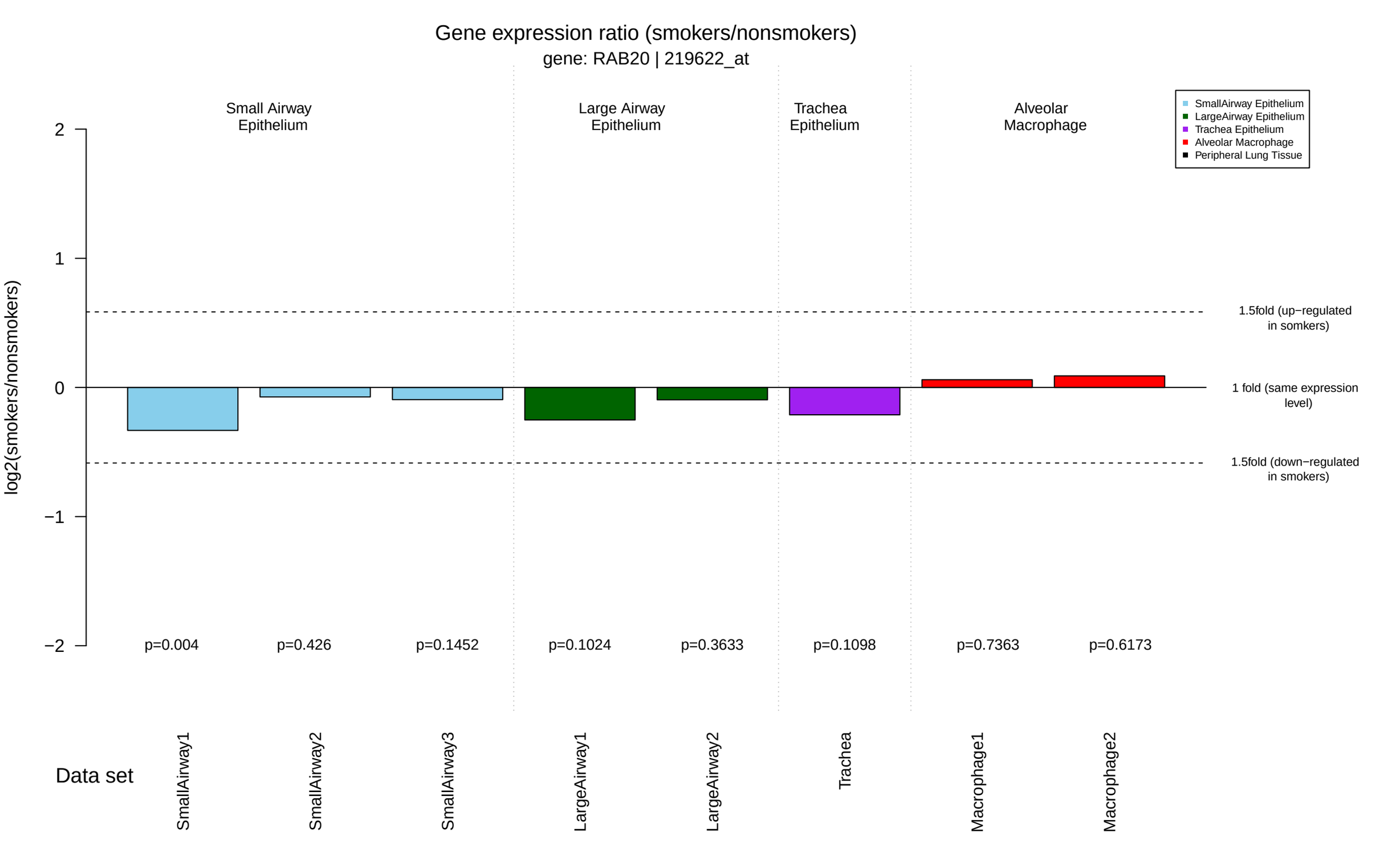
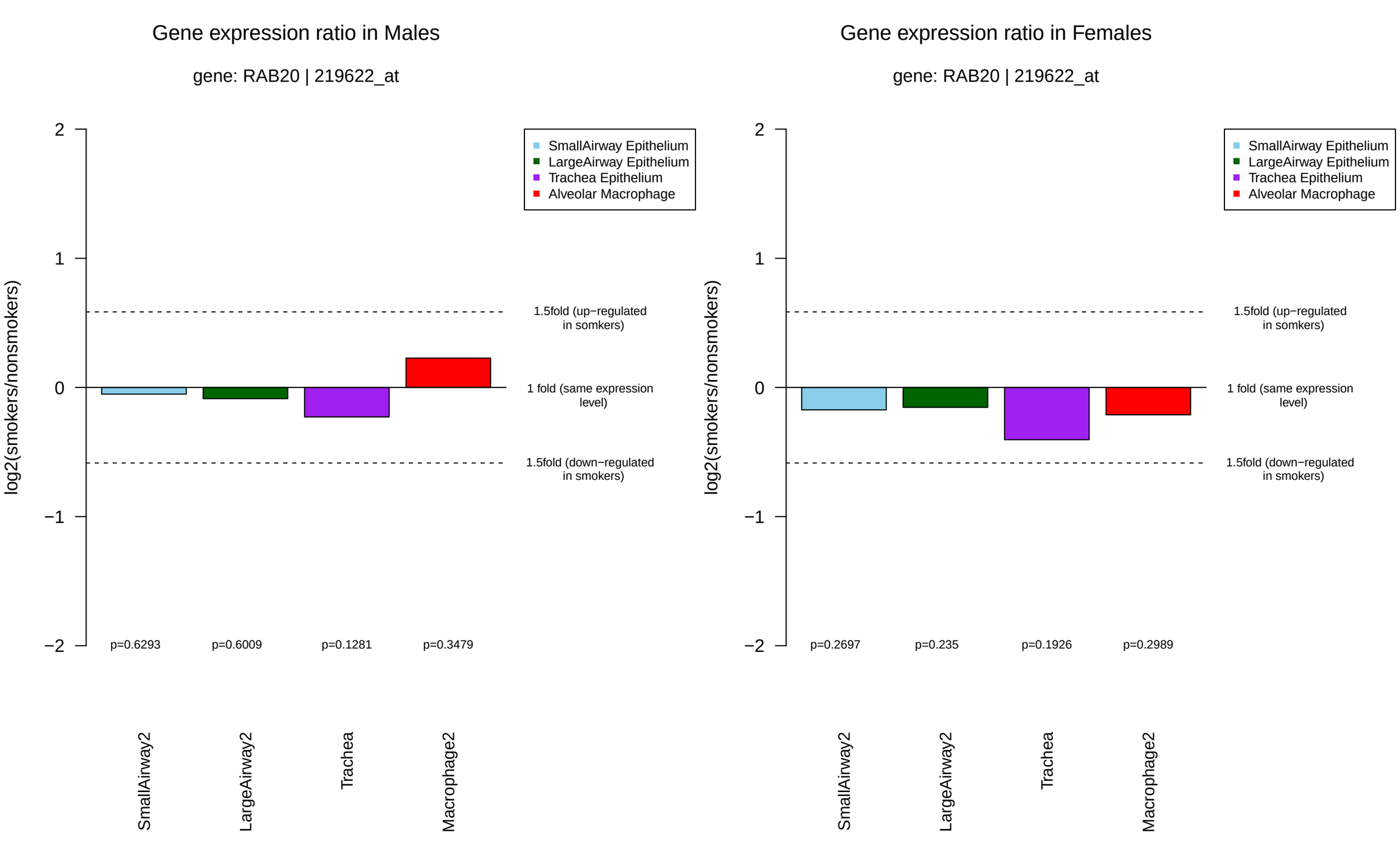
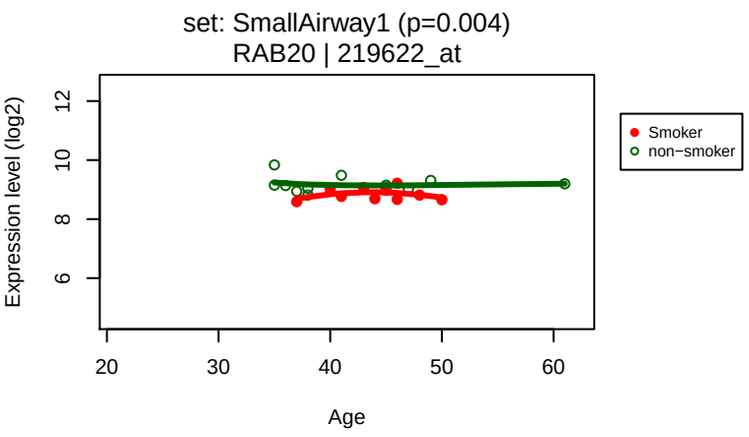
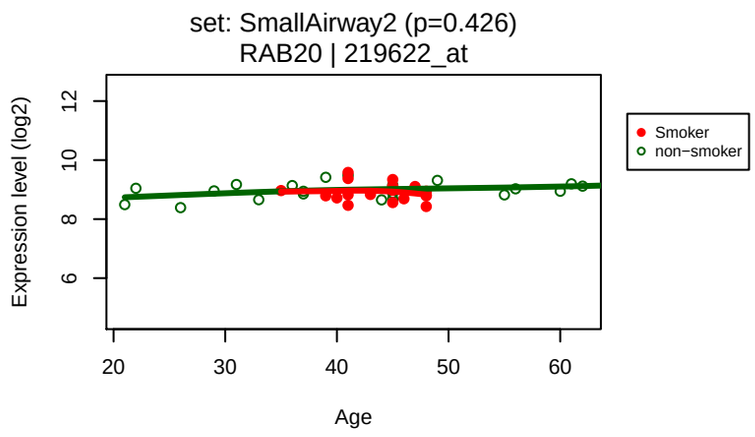
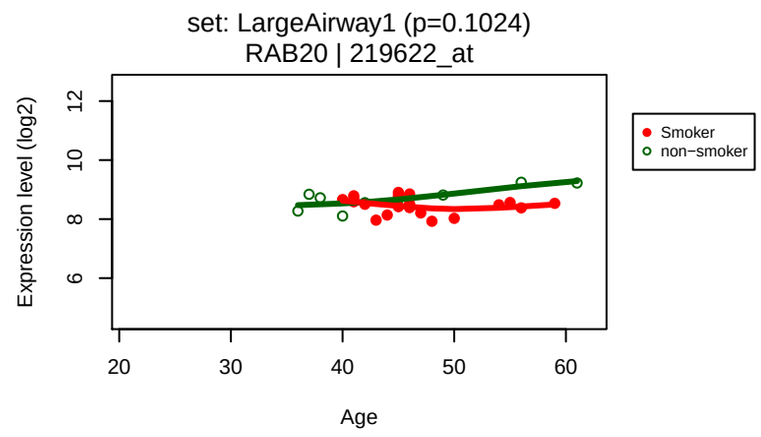

Gene expression of smokers and nonsmokers
gene: RAB20 | 219622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.004
p=0.426
p=0.1452
p=0.1024
p=0.3633
p=0.1098
p=0.7363
p=0.6173
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

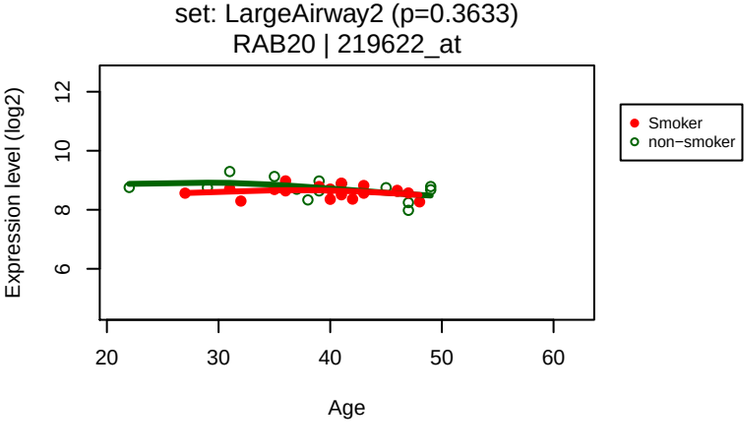

Gene expression ratio (smokers/nonsmokers)
gene: RAB20 | 219622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.004
p=0.426
p=0.1452
p=0.1024
p=0.3633
p=0.1098
p=0.7363
p=0.6173
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

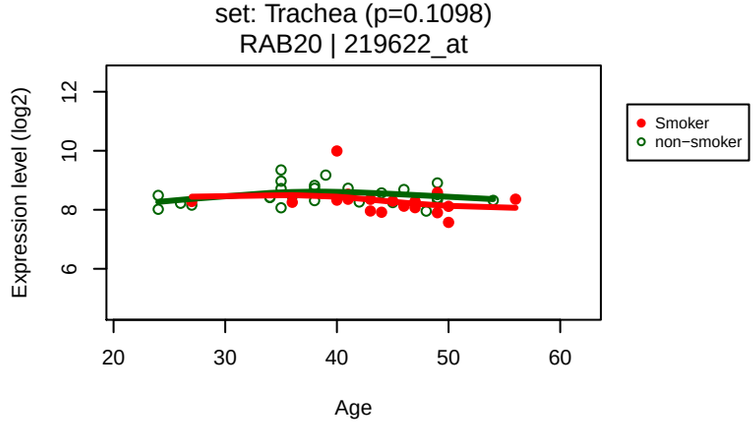

Gene expression ratio in Males
gene: RAB20 | 219622_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6293
p=0.6009
p=0.1281
p=0.3479
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB20 | 219622_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2697
p=0.235
p=0.1926
p=0.2989
SmallAirway2
LargeAirway2
Trachea
Macrophage2

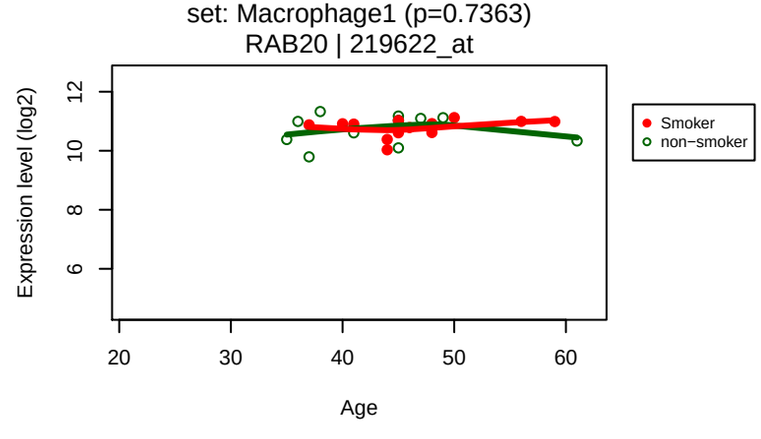

set: SmallAirway1 (p=0.004)
RAB20 | 219622_at
Expression level (log2)
Age
Smoker
non−smoker

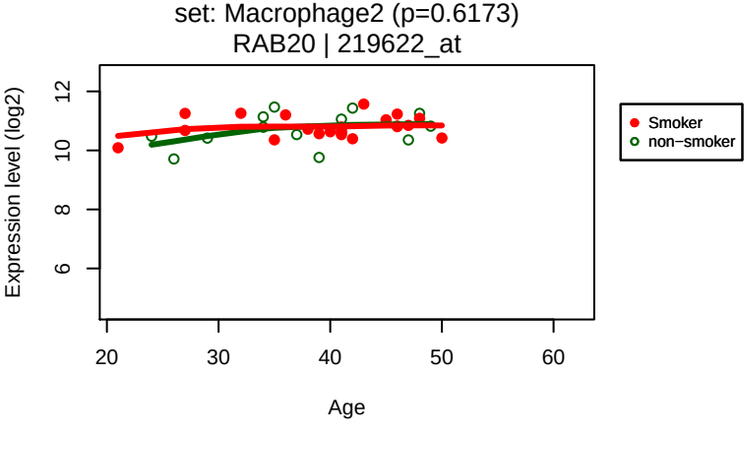

set: SmallAirway2 (p=0.426)
RAB20 | 219622_at
Expression level (log2)
Age
Smoker
non−smoker

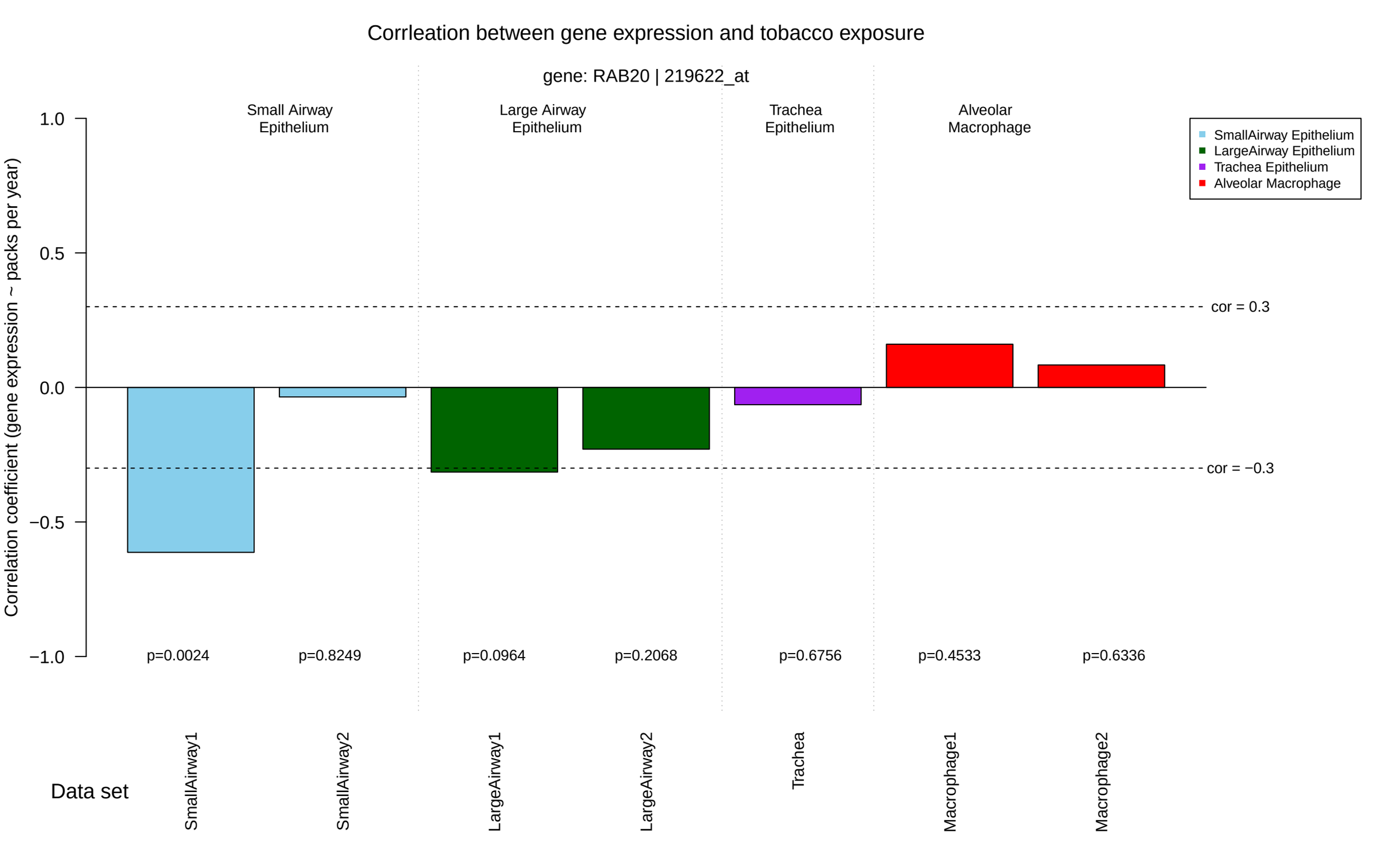

set: LargeAirway1 (p=0.1024)
RAB20 | 219622_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3633)
RAB20 | 219622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1098)
RAB20 | 219622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7363)
RAB20 | 219622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6173)
RAB20 | 219622_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB20 | 219622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0024
p=0.8249
p=0.0964
p=0.2068
p=0.6756
p=0.4533
p=0.6336
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2